

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent
Kind Code
Date of Patent
Inventor(s)

12410050
B2
September 09, 2025
Jin; Changdao

Dumping-type autonomous quantitative liquid discharging device and quantitative liquid discharging method thereof

Abstract

The present invention relates to a dumping-type automatic quantitative liquid discharging device, which comprises an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel; the second liquid storage cavity further comprises a liquid storage part; the liquid storage part captures and stores a part of liquid to become a new quantitative liquid; when the device is reset, the quantitative liquid stored in the second liquid storage cavity will be received and stored by the first liquid storage cavity; and a quantitative liquid discharging method, which comprises the following steps: providing a dumping-type automatic quantitative liquid discharging device, and pouring the liquid into an accommodating cavity; connecting a device cover with the accommodating cavity; tilting the dumping-type automatic quantitative liquid discharging device to allow quantitative liquid to flow out.

Inventors: Jin; Changdao (Ningbo, CN)
Applicant: Jin; Changdao (Ningbo, CN)
Family ID: 95861846
Appl. No.: 18/527422
Filed: December 04, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20250178881 A1	Jun. 05, 2025

Publication Classification

Int. Cl.: **B67D3/00** (20060101); **A47G19/18** (20060101); **B65D25/28** (20060101); **B65D47/24** (20060101); **B65D47/32** (20060101)

U.S. Cl.:

CPC **B67D3/0045** (20130101); **A47G19/18** (20130101); **B65D25/2811** (20130101); **B65D47/245** (20130101); **B65D47/32** (20130101); **B67D3/0051** (20130101);

Field of Classification Search

CPC: B65D (47/26); B65D (47/32); B65D (47/245); B67D (3/0041); B67D (3/0045); B67D (3/0051); A47G (19/18)

USPC: 222/36; 222/56

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
6409046	12/2001	Peckels	222/37	B67D 3/0077
8985406	12/2014	Tachi	222/472	B65D 59/02
9359183	12/2015	Henderson	N/A	B67D 3/0051
12110157	12/2023	Tu	N/A	B65D 51/242

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
WO-2020006600	12/2019	WO	B67D 3/0045

Primary Examiner: Maust; Timothy L

Background/Summary

TECHNICAL FIELD

(1) The present invention relates to the technical field of kitchen utensils, in particular to a dumping-type automatic quantitative liquid discharging device and a quantitative liquid discharging method thereof.

BACKGROUND

(2) Container devices are indispensable items in human life. They can store, transport, protect and display various substances to meet people's various needs. The types and materials of devices are very rich, including metal, plastic, glass, ceramics, paper and so on.

(3) However, the existing container devices in the market generally have the problem that it is difficult to control the liquid output. The unstable liquid output will lead to the inaccurate liquid quantity filled or transported, which will affect the consistency and efficiency of products. It leads to waste or shortage of liquid, which affects the quality and cost of products. For example, a container device disclosed in U.S. Patent No. 20230278761 has the problem that it is difficult to control the liquid output; Similarly, the U.S. patent with the application Ser. No. 13/907,526 also has unstable liquid discharging.

(4) Therefore, it is necessary for us to put forward a dumping-type automatic quantitative liquid

discharging device, which has stable liquid outlet, stable structure and convenient use, and can provide users with the choice of devices with stable liquid outlet and obtain quantitative liquid more conveniently.

SUMMARY

(5) The terms “invention,” “the invention,” “this invention” and “the present invention” used in this patent are intended to refer broadly to all of the subject matter of this patent and the patent claims below. Statements containing these terms should be understood not to limit the subject matter described herein or to limit the meaning or scope of the patent claims below. Embodiments of the invention covered by this patent are defined by the claims below, not this summary. This summary is a high-level overview of various embodiments of the invention and introduces some of the concepts that are further described in the Detailed Description section below. This summary is not intended to identify key or essential features of the claimed subject matter, nor is it intended to be used in isolation to determine the scope of the claimed subject matter. The subject matter should be understood by reference to appropriate portions of the entire specification of this patent, any or all drawings and each claim.

(6) The present invention provides a dumping-type automatic quantitative liquid discharging device, which includes: an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel;

(7) wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and

(8) wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; when said device pours out liquid, the quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid channel, and an atmospheric low pressure area is formed at the other end of said accommodating cavity; and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is confined between said first liquid storage cavity and said accommodating cavity; and

(9) wherein said second liquid storage cavity captures and stores a part of said liquid to become new said quantitative liquid; and

(10) wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity.

(11) The present invention also provides a dumping-type automatic quantitative liquid discharging device, which includes an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel;

(12) wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and

(13) wherein a sum of volumes of said first liquid storage cavity and said second liquid storage cavity is smaller than a volume of said accommodating cavity, and said second liquid storage cavity further comprises a liquid storage part; and

(14) wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; when said device pours out liquid, said quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid channel; and an atmospheric low pressure area is formed at the other end of said accommodating cavity, and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is confined between said first liquid storage cavity and said accommodating cavity; and

(15) wherein said liquid storage part captures and stores a part of said liquid to become new said quantitative liquid; and

- (16) wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity.
- (17) The present invention also provides a quantitative liquid discharging method, which includes providing a dumping-type automatic quantitative liquid discharging device, wherein dumping-type automatic quantitative liquid discharging device comprises an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel;
- (18) wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and
- (19) wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; when said device pours out liquid, the quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid channel, and an atmospheric low pressure area is formed at the other end of said accommodating cavity; and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is confined between said first liquid storage cavity and said accommodating cavity; and
- (20) wherein said second liquid storage cavity captures and stores a part of said liquid to become new said quantitative liquid; and
- (21) wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity; and
- (22) pouring liquid into said accommodating cavity; and
- (23) connecting said device cover with said accommodating cavity; and
- (24) tilting said dumping-type automatic quantitative liquid discharging device to allow the quantitative liquid to flow out.
-

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) In order to explain the technical scheme of this application more clearly, the drawings needed in the implementation will be briefly introduced below. Obviously, the drawings described below are only some implementations of this application. For those skilled in the art, other drawings can be obtained according to these drawings without creative work.
- (2) FIG. 1 is a schematic diagram of a dumping-type automatic quantitative liquid discharging device;
- (3) FIG. 2 is an explosion diagram of the dumping-type automatic quantitative liquid discharging device;
- (4) FIG. 3 is an enlarged schematic view at O in FIG. 2;
- (5) FIG. 4 is an exploded view of the device cover;
- (6) FIG. 5 is another exploded view of the device cover;
- (7) FIG. 6 is a schematic bottom view of the device cover;
- (8) FIG. 7 is a schematic diagram of the device when it is inverted;
- (9) FIG. 8 is a schematic diagram of the device when it is put in the upright position;
- (10) FIG. 9 is a schematic diagram of another embodiment of the dumping-type automatic quantitative liquid discharging device.
- (11) In the drawings: **1000**, Device; **1001**, Accommodating cavity; **1002**, Bottom groove; **2000**, Device cover; **2001**, Fluid channel; **2002**, Liquid outlet; **3002**, First liquid storage cavity; **3003**, Second liquid storage cavity; **3004**, Liquid inlet hole; **3005**, Liquid storage cavity; **3006**, Liquid storage outlet; **3007**, First valve body; **3008**, First sealing ring; **3009**, Sealing opening; **3010**,

Embedding groove; **3011**, First clamping groove; **3012**, First clamping block; **3013**, Liquid retaining ring; **3014**, Liquid retaining part; **3015**, Clamping ring; **3016**, Ring groove; **3017**, Top cover; **3018**, Rotating column; **3019**, Rotating groove; **3020**, Weight block; **3021**, Accommodating part; **3022**, Liquid outlet nozzle; **3023**, Cover nozzle; **3024**, First fixing groove; **3025**, Second fixing groove; **3026**, Fixing ring; **3027**, Air inlet pipe; **3028**, Second clamping block; **3029**, Second clamping groove; **3030**, Handle part; **3031**, Anti-skid lines; **3032**, Limiting edge; **3033**, External thread; **3034**, Internal thread; **3035**, Second sealing ring; **3036**, Sealing groove; **3037**, Liquid storage; **3038**, Air outlet hole; **4000**, Second valve body; A, Liquid; B, Quantitative liquid; C, Atmospheric low pressure area; D, Atmospheric normal pressure area.

DESCRIPTION OF EMBODIMENTS

(12) In describing the preferred embodiments, specific terminology will be resorted to for the sake of clarity. It is to be understood that each specific term includes all technical equivalents which operate in a similar manner to accomplish a similar purpose.

(13) While various aspects and features of certain embodiments have been summarized above, the following detailed description illustrates a few exemplary embodiments in further detail to enable one skilled in the art to practice such embodiments. Reference will now be made in detail to embodiments of the inventive concept, examples of which are illustrated in the accompanying drawings. The accompanying drawings are not necessarily drawn to scale. The described examples are provided for illustrative purposes and are not intended to limit the scope of the invention. It should be understood, however, that persons having ordinary skill in the art may practice the inventive concept without these specific details.

(14) It will be understood that, although the terms first, second, etc. may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another. For example, a first attachment could be termed a second attachment, and, similarly, a second attachment could be termed a first attachment, without departing from the scope of the inventive concept.

(15) It will be understood that when an element or layer is referred to as being “on,” “coupled to,” or “connected to” another element or layer, it can be directly on, directly coupled to or directly connected to the other element or layer, or intervening elements or layers may be present. In contrast, when an element is referred to as being “directly on,” “directly coupled to,” or “directly connected to” another element or layer, there are no intervening elements or layers present. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items.

(16) As used in the description of the inventive concept and the appended claims, the singular forms “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates other.

(17) As shown in FIGS. **1** to **9**, the present invention provides a dumping-type automatic quantitative liquid discharging device, which includes an accommodating cavity **1001** for accommodating liquid A, a first liquid storage cavity **3002** and a second liquid storage cavity **3003** for storing quantitative liquid B, and a fluid channel **2001**.

(18) The fluid channel **2001** communicates with the accommodating cavity **1001**, the first liquid storage cavity **3002** and the second liquid storage cavity **3003** in turn, and defines the path for air and the quantitative liquid B to enter and leave the device **1000**.

(19) The sum of the volumes of the first liquid storage cavity **3002** and the second liquid storage cavity **3003** is smaller than the volume of the accommodating cavity **1001**, and the second liquid storage cavity **3003** further includes a liquid storage part **3037**.

(20) The first liquid storage cavity **3002** and the second liquid storage cavity **3003** are arranged at one end of the accommodating cavity **1001**. When the device **1000** pours out liquid, the quantitative liquid B stored in the second liquid storage cavity **3003** leaves the device **1000** through the fluid channel **2001**, and an atmospheric low pressure area C is formed at the other end of the

accommodating cavity **1001**, and an atmospheric normal pressure area D is formed in the first liquid storage cavity **3002** and the second liquid storage cavity **3003**.

(21) The liquid storage part **3037** captures and stores a part of the liquid A to become a new quantitative liquid B.

(22) When the device **1000** is reset, the quantitative liquid B stored in the second liquid storage cavity **3003** will be received and stored by the first liquid storage cavity **3002**.

(23) As shown in FIGS. 2 and 3, in this embodiment, the second liquid storage cavity **3003** is provided with a liquid storage outlet **3006** corresponding to the fluid channel **2001**, and a first valve body **3007** corresponding to the liquid storage outlet **3006** is provided in the fluid channel **2001**.

When the device is in a vertical state, the first valve body **3007** blocks the liquid storage outlet **3006**; when the device pours out liquid, the first valve body **3007** is far away from the liquid storage outlet **3006**; the liquid storage outlet **3006** is provided with a first sealing ring **3008**, and the first sealing ring **3008** is provided with a sealing opening **3009** corresponding to the first valve body **3007**. The first valve body **3007** improves the air tightness of the device, and can prevent the liquid in the device from volatilizing and deteriorating.

(24) The second liquid storage cavity **3003** is provided with a liquid inlet hole **3004** for allowing the liquid A to enter the second liquid storage cavity **3003**, and an air outlet hole **3038** for discharging the air in the second liquid storage cavity **3003**. This design can make the device back to the right time, and the liquid in the liquid storage part **3037** can smoothly enter the second liquid storage cavity **3003**.

(25) In this embodiment, the liquid inlet hole **3004** and the air outlet hole **3038** are provided as one. In other embodiments (not shown in the figures), the liquid inlet hole **3004** and the air outlet hole **3038** are not limited to one, but can also be set to two, three, four, five and any desired number.

(26) In this embodiment, the first valve body **3007** is provided as a sphere. In other embodiments (not shown in the figures), the first valve body is not limited to being set as a sphere, but can also be set as a cone, a cylinder, a cube, a cuboid, an ellipsoid and any desired shape.

(27) In this embodiment, the first sealing ring **3008** is made of a rubber material. In other embodiments, the first sealing ring **3008** is not limited to being made of rubber, but can also be made of polytetrafluoroethylene (PTFE), polyurethane (PU), metal and any other desired materials. In some embodiments, the first sealing ring **3008** is not provided between the fluid channel **2001** and the liquid storage outlet **3006** of the second liquid storage cavity **3003**, but the fluid channel **2001** or the second liquid storage cavity **3003** is made of the material of the first sealing ring **3008**.

(28) Rubber is the most commonly used sealing ring material with good elasticity, softness, wear resistance and chemical resistance. There are many kinds of rubber, and different rubbers have different properties and application scope, such as oil resistance of NBR, high and low temperature resistance of silicone rubber, heat resistance of fluororubber, acid and alkali, etc.

(29) Polytetrafluoroethylene (PTFE), a synthetic polymer material, has very low friction coefficient, high temperature resistance, excellent chemical stability and non-stickiness. PTFE can tolerate a variety of media, such as oil, water, steam, medicine, etc., and is suitable for high temperature, high speed and high pressure sealed environment.

(30) Polyurethane (PU) is a synthetic elastomer material with high strength, hardness, wear resistance and aging resistance. PU can tolerate oil, water, oxygen and oarea, and is suitable for high-pressure and high-speed system sealing in construction machinery and metallurgical equipment.

(31) Metal is a hard sealing ring material with high mechanical strength, heat resistance and corrosion resistance. There are many kinds of metals, and different metals have different properties and scope of application, such as steel, iron, copper and stainless steel. Metal sealing ring is generally used in high pressure, high temperature and high speed sealing environment, or combined with other materials.

(32) As shown in FIGS. 2 and 3, in this embodiment, an embedding groove **3010** is arranged at the

bottom of the fluid channel **2001**, and the liquid storage outlet **3006** of the second liquid storage cavity **3003** and the first sealing ring **3008** are arranged in the embedding groove **3010**; the embedding groove **3010** is provided with a first clamping groove **3011**, and the second liquid storage cavity **3003** is provided with a first clamping block **3012** corresponding to the first clamping groove **3011**, so that the fluid channel **2001** and the first liquid storage cavity **3002** are detachably connected by clamping.

(33) In this embodiment, the fluid channel **2001** and the first liquid storage cavity **3002** are detachably connected by clamping. In other embodiments (not shown in the figures), the fluid channel and the first liquid storage cavity are not limited to being detachably connected by clamping, but can also be set to be connected by screws, pins, rivets, welding and any desired connection mode.

(34) As shown in FIGS. 2 to 4, in this embodiment, the first liquid storage cavity **3002** is provided with a liquid retaining ring **3013**, and the liquid retaining ring **3013** is formed with a liquid retaining part **3014** bent inward; the first liquid storage cavity **3002** is provided with a clamping ring **3015**, and the liquid retaining ring **3013** is provided with a ring groove **3016**. The first liquid storage cavity **3002** and the liquid retaining ring **3013** are detachably connected by clamping.

(35) In this embodiment, the first liquid storage cavity **3002** and the liquid retaining ring **3013** are detachably connected by clamping. In other embodiments (not shown in the figures), the first liquid storage cavity and the liquid storage retaining ring are not limited to being detachably connected by clamping, but can also be connected by screws, pins, rivets, welding and any other desirable connection methods.

(36) As shown in FIG. 5, in this embodiment, a top cover **3017** is provided on the device cover **2000**, a rotating column **3018** is provided on the device cover **2000**, a rotating groove **3019** corresponding to the rotating column **3018** is provided on the top cover **3017**, and the top cover **3017** is rotatably connected with the device cover **2000**.

(37) In this embodiment, the top cover **3017** and the device cover **2000** are detachably connected by clamping. In other embodiments (not shown in the figures), the top cover and the device cover are not limited to the detachable connection by clamping, but can also be set as screw connection, pin connection, rivet connection and any desired connection mode.

(38) In this embodiment, the top cover **3017** is provided with a weight block **3020**, wherein the top cover **3017** is formed with a receiving part **3021**, and the weight block **3020** is arranged in the receiving part **3021**. When the device tilts to pour out liquid, the end of the top cover **3017** with the weight block **3020** sinks due to the traction of the weight block **3020**, and the end of the top cover **3017** far away from the weight block **3020** tilts up, thus realizing the automatic opening of the top cover **3017** when the device pours out liquid.

(39) As shown in FIG. 5, in this embodiment, the device cover **2000** is provided with a liquid outlet **2002** corresponding to the fluid channel and a liquid outlet nozzle **3022** corresponding to the liquid outlet **2002**, which communicates with the fluid channel, and a cover nozzle **3023** corresponding to the liquid outlet nozzle **3022** is formed on the device cover **2000**; a first fixing groove **3024** is arranged on the liquid outlet nozzle **3022**, a second fixing groove **3025** corresponding to the first fixing groove **3024** is arranged on the device cover **2000**, and a fixing ring **3026** corresponding to the first fixing groove **3024** and the second fixing groove **3025** is arranged between the liquid outlet nozzle **3022** and the device cover **2000**; and the liquid outlet nozzle **3022** and the device cover **2000** are detachably connected through the first fixing groove **3024**, the second fixing groove **3025** and the fixing ring **3026**.

(40) In this embodiment, the liquid outlet nozzle **3022** and the device cover **2000** are detachably connected through the first fixing groove **3024**, the second fixing groove **3025** and the fixing ring **3026**. In other embodiments (not shown in the figures), the liquid outlet nozzle and the device cover are not limited to being detachably connected by the first fixing groove, the second fixing groove and the fixing ring, but also can be connected by screws, pins, rivets, welding and any other

desired connection methods.

(41) As shown in FIGS. 2 to 5, in this embodiment, an air inlet pipe **3027** is provided on the device cover **2000**, wherein the air inlet pipe **3027** communicates with the fluid channel **2001**. The air inlet pipe **3027** can balance the internal pressure of the device when pouring liquid, making the liquid discharging process smoother.

(42) As shown in FIGS. 2 and 3, in this embodiment, the second liquid storage cavity **3003** is provided with a second clamping block **3028**, and the first liquid storage cavity **3002** is provided with a second clamping groove **3029** corresponding to the second clamping block **3028**, so that the first liquid storage cavity **3002** and the second liquid storage cavity **3003** are detachably connected by clamping.

(43) In this embodiment, the first liquid storage cavity **3002** and the second liquid storage cavity **3003** are detachably connected by clamping. In other embodiments (not shown in the figures), the first liquid storage cavity and the second liquid storage cavity are not limited to being detachably connected by clamping, but can also be connected by screws, pins, rivets, welding and any other desired connection methods.

(44) As shown in FIG. 1, in this embodiment, a handle **3030** is provided on the device cover **2000**. In some embodiments, the handle **3030** is also provided with an anti-skid pad, anti-skid lines and a shape that conforms to the grip of a human hand.

(45) As shown in FIG. 1, in this embodiment, the accommodating cavity **1001** has an outer surface corresponding to the inner surface, and anti-skid lines **3031** are provided on the outer surface. In other embodiments (not shown in the figures), the outer surface is not limited to being provided with anti-skid lines, but also provided with scales, patterns and anti-skid pads.

(46) As shown in FIGS. 2 and 3, in this embodiment, at least one limiting edge **3032** corresponding to the first valve body **3007** is provided in the fluid channel **2001**. The limiting edge **3032** is used to prevent the first valve body **3007** from blocking the liquid outlet **2002**, so that the liquid outlet process is smoother.

(47) As shown in FIGS. 2 to 6, in this embodiment, an external thread **3033** is provided on the accommodating cavity **1001** and an internal thread **3034** is provided on the device cover **2000**, and the accommodating cavity **1001** and the device cover **2000** are detachably connected by threads.

(48) In this embodiment, the accommodating cavity **1001** and the device cover **2000** are detachably connected by threads. In other embodiments (not shown in the figures), the accommodating cavity and the device cover are not limited to being detachably connected by threads, but can also be arranged as screw connection, pin connection, rivet connection, welding, snap connection and any desired connection mode.

(49) As shown in FIGS. 4 to 6, in this embodiment, the device cover **2000** is provided with a second sealing ring **3035** and a sealing groove **3036** corresponding to the second sealing ring **3035**. The arrangement of the second sealing ring **3035** increases the sealing property of the device, so that the device **1000** is closed relative to the outside, and only the fluid channel is a channel communicating with the outside.

(50) As shown in FIG. 2, in this embodiment, the bottom of the accommodating cavity **1001** is provided with a bottom groove **1002**, which can help the device to keep stable. When the surface on which the device is placed is uneven, the periphery of the bottom groove **1002** contacts the ground, thus reducing the contact area with the uneven surface.

(51) As shown in FIGS. 7 to 8, when the device **1000** is placed upright, the accommodating cavity **1001** is only in air communication with the first liquid storage cavity **3002**, and the air pressure of the two cavities is the same. The first liquid storage cavity **3002** and the second liquid storage cavity **3003** are connected by a liquid inlet hole **3004** to form a communication device structure, and the liquid levels of the two cavities are equal.

(52) When the device **1000** is inverted, the quantitative liquid B in the second liquid storage cavity **3003** flows out under the action of gravity; the liquid in the second liquid storage cavity **3003** falls;

the second liquid storage cavity **3003** is in air communication with the first liquid storage cavity **3002**, and the air pressure is the same; the liquid in the accommodating cavity **1001** falls down and forms a communication structure with the second liquid storage cavity **3003**. According to the communicating vessel principle, if the air pressures of the accommodating cavity **1001** and the second liquid storage cavity **3003** are equal, the liquid levels of the accommodating cavity **1001** and the second liquid storage cavity **3003** will tend to be equal. However, at this time, the accommodating cavity **1001** is closed, and the liquid in the accommodating cavity **1001** is subjected to gravity, so that the air pressure above the liquid level of the accommodating cavity **1001** is increased. When the gravity of the liquid in the accommodating cavity **1001** is balanced with the atmospheric pressure of the second liquid storage cavity **3003**, the liquid levels in the accommodating cavity **1001** and the second liquid storage cavity **3003** are balanced, and the liquid in the accommodating cavity **1001** will not flow out through the second liquid storage cavity **3003** and the first liquid storage cavity **3002**.

(53) When the device **1000** is restored to the normal state, the liquid A in the accommodating cavity **1001** falls back to the cavity bottom, and due to the atmospheric pressure, air enters the second liquid storage cavity **3003** and the accommodating cavity **1001** from the first liquid storage cavity **3002** through the air outlet hole **3038**; in addition, due to the action of gravity, the liquid in the second liquid storage cavity **3003** falls back, and part of it enters the first liquid storage cavity **3002** through the liquid inlet hole **3004**, forming a communicating vessel structure.

(54) As shown in FIG. 9, it is another embodiment of the dumping-type automatic quantitative liquid discharging device provided by the present invention, which is different from the above-mentioned embodiment in that a second valve body **4000** is arranged in the second liquid storage cavity **3003**, which is rollably arranged in the second liquid storage cavity **3003** and closely adheres to the second liquid storage cavity **3003** for blocking; when the device is tilted to discharge the liquid in the second liquid storage cavity **3003** through the fluid channel **2001**, the second valve body **4000** is used to abut against the inner wall surface of the upper end of the second liquid storage cavity **3003** to block the liquid storage outlet **3006**.

(55) In this embodiment, the second valve body **4000** further improves the airtightness of the device, and when the device pours liquid, the liquid in the device **1000** will not communicate with the outside world, further preventing volatilization and deterioration.

(56) The present invention provides a quantitative liquid discharging method, which includes providing a dumping-type automatic quantitative liquid discharging device, wherein dumping-type automatic quantitative liquid discharging device comprises an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel;

(57) wherein, the fluid channel sequentially communicates with the accommodating cavity, the first liquid storage cavity and the second liquid storage cavity, and defines a path for air and the quantitative liquid to enter and leave the device; and

(58) wherein, the first liquid storage cavity and the second liquid storage cavity are arranged at one end of the accommodating cavity; when the device pours out liquid, the quantitative liquid stored in the second liquid storage cavity leaves the device through the fluid channel, and an atmospheric low pressure area is formed at the other end of the accommodating cavity; and an atmospheric normal pressure area is formed in the first liquid storage cavity and the second liquid storage cavity, so that the liquid is confined between the first liquid storage cavity and the accommodating cavity; and

(59) wherein the second liquid storage cavity captures and stores a part of the liquid to become new the quantitative liquid; and

(60) wherein, when the device is reset, the quantitative liquid stored in the second liquid storage cavity will be received and stored by the first liquid storage cavity; and

(61) pouring liquid into the accommodating cavity; and

(62) connecting the device cover with the accommodating cavity; and

(63) tilting the dumping-type automatic quantitative liquid discharging device to allow the quantitative liquid to flow out.

(64) The terms “comprising,” “including,” “having,” and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth. Also, the term “or” is used in its inclusive sense (and not in its exclusive sense) so that when used, for example, to connect a list of elements, the term “or” means one, some, or all of the elements in the list. The use of “adapted to” or “configured to” herein is meant as open and inclusive language that does not foreclose devices adapted to or configured to perform additional tasks or steps. Additionally, the use of “based on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Similarly, the use of “based at least in part on” is meant to be open and inclusive, in that a process, step, calculation, or other action “based at least in part on” one or more recited conditions or values may, in practice, be based on additional conditions or values beyond those recited. Headings, lists, and numbering included herein are for ease of explanation only and are not meant to be limiting.

(65) The various features and processes described above may be used independently of one another, or may be combined in various ways. All possible combinations and sub-combinations are intended to fall within the scope of the present disclosure. In addition, certain method or process blocks may be omitted in some implementations. The methods and processes described herein are also not limited to any particular sequence, and the blocks or states relating thereto can be performed in other sequences that are appropriate. For example, described blocks or states may be performed in an order other than that specifically disclosed, or multiple blocks or states may be combined in a single block or state. The example blocks or states may be performed in serial, in parallel, or in some other manner. Blocks or states may be added to or removed from the disclosed examples. Similarly, the example systems and components described herein may be configured differently than described. For example, elements may be added to, removed from, or rearranged compared to the disclosed examples.

(66) The invention has now been described in detail for the purposes of clarity and understanding. However, those skilled in the art will appreciate that certain changes and modifications may be practiced within the scope of the appended claims.

(67) Conditional language used herein, such as, among others, “can,” “could,” “might,” “may,” “e.g.,” and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain examples include, while other examples do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements and/or steps are in any way required for one or more examples or that one or more examples necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular example.

Claims

1. A dumping-type automatic quantitative liquid discharging device, comprising: an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel; and wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; and when said device pours out liquid, the quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid

channel, and an atmospheric low pressure area is formed at the other end of said accommodating cavity; and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is confined between said first liquid storage cavity and said accommodating cavity; and wherein said second liquid storage cavity captures and stores a part of said liquid to become new said quantitative liquid; and wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity.

2. The dumping-type automatic quantitative liquid discharging device according to claim 1, wherein a sum of volumes of said first liquid storage cavity and said second liquid storage cavity is smaller than a volume of said accommodating cavity, and said second liquid storage cavity further comprises a liquid storage part which captures and stores a part of said liquid to become new said quantitative liquid; and said second liquid storage cavity further comprises a liquid storage outlet, and a first valve body corresponding to said liquid storage outlet is arranged in said fluid channel, and when said device is in a vertical state, said first valve body blocks said liquid storage outlet; and when said device pours out liquid, said first valve body is far away from said liquid storage outlet; and wherein, a first sealing ring is arranged on the liquid storage outlet, and a sealing opening corresponding to said first valve body is arranged on said first sealing ring.

3. The dumping-type automatic quantitative liquid discharging device according to claim 2, wherein a second valve body is arranged in said second liquid storage cavity, and said second valve body is rollably arranged in said second liquid storage cavity and closely adheres to said second liquid storage cavity for blocking.

4. The dumping-type automatic quantitative liquid discharging device according to claim 1, wherein said first liquid storage cavity is provided with a liquid storage retaining ring, and said liquid storage retaining ring is formed with a liquid retaining part bent inward; and wherein, a clamping ring is arranged on said first liquid storage cavity, a ring groove is arranged on said liquid storage retaining ring, and said first liquid storage cavity and said liquid storage retaining ring are detachably connected by clamping.

5. The dumping-type automatic quantitative liquid discharging device according to claim 2, wherein an embedding groove is arranged at a bottom of said fluid channel, and the liquid storage outlet of said second liquid storage cavity and said first sealing ring are arranged in said embedding groove, and wherein, said embedding groove is provided with a first clamping groove, said second liquid storage cavity is provided with a first clamping block corresponding to said first clamping groove, and said fluid channel and said first liquid storage cavity are detachably connected by clamping; and wherein, a second clamping block is arranged on said second liquid storage cavity, a second clamping groove corresponding to said second clamping block is arranged on said first liquid storage cavity, and said first liquid storage cavity and said second liquid storage cavity are detachably connected by clamping.

6. A dumping-type automatic quantitative liquid discharging device, comprising: an accommodating cavity for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel; and wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and wherein a sum of volumes of said first liquid storage cavity and said second liquid storage cavity is smaller than a volume of said accommodating cavity, and said second liquid storage cavity further comprises a liquid storage part; and wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; and when said device pours out liquid, said quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid channel; and an atmospheric low pressure area is formed at the other end of said accommodating cavity, and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is

confined between said first liquid storage cavity and said accommodating cavity; and wherein said liquid storage part captures and stores a part of said liquid to become new said quantitative liquid; and wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity.

7. The dumping-type automatic quantitative liquid discharging device according to claim 6, wherein said second liquid storage cavity further comprises a liquid storage outlet, and a first valve body corresponding to said liquid storage outlet is arranged in said fluid channel; and when said device is in a vertical state, said first valve body blocks said liquid storage outlet; and when said device pours liquid, said first valve body is far away from said liquid storage outlet, wherein a first sealing ring is arranged on said liquid storage outlet, and a sealing opening corresponding to said first valve body is arranged on said first sealing ring; and wherein said second liquid storage cavity is provided with a liquid inlet hole and an air outlet hole.

8. The dumping-type automatic quantitative liquid discharging device according to claim 7, wherein an embedding groove is arranged at a bottom of said fluid channel, and said liquid storage outlet of said second liquid storage cavity and said first sealing ring are arranged in said embedding groove; and wherein, said embedding groove is provided with a first clamping groove, said second liquid storage cavity is provided with a first clamping block corresponding to said first clamping groove, and said fluid channel and said first liquid storage cavity are detachably connected by clamping.

9. The dumping-type automatic quantitative liquid discharging device according to claim 6, wherein said first liquid storage cavity is provided with a liquid storage retaining ring, and said liquid storage retaining ring is formed with a liquid retaining part bent inward; and wherein, a clamping ring is arranged on said first liquid storage cavity, a ring groove is arranged on said liquid storage retaining ring, and said first liquid storage cavity and said liquid storage retaining ring are detachably connected by clamping.

10. The dumping-type automatic quantitative liquid discharging device according to claim 6, wherein said device comprises a device cover fixedly connected with said accommodating cavity, and said device cover is provided with a top cover, a rotating column is arranged on said device cover, and a rotating groove corresponding to said rotating column is arranged on said top cover, and said top cover is rotatably connected with said device cover; and wherein, a weight block is arranged on said top cover, wherein said top cover is formed with an accommodating part, and said weight block is arranged in said accommodating part.

11. The dumping-type automatic quantitative liquid discharging device according to claim 10, wherein a liquid outlet corresponding to said fluid channel and a liquid outlet nozzle corresponding to said liquid outlet are arranged on said device cover, and said liquid outlet nozzle is communicated with said fluid channel, and a cover nozzle corresponding to said liquid outlet nozzle is formed on said device cover; and wherein, said liquid outlet nozzle is provided with a first fixing groove, said device cover is provided with a second fixing groove corresponding to said first fixing groove, and a fixing ring corresponding to said first fixing groove and said second fixing groove is arranged between said liquid outlet nozzle and said device cover; and said liquid outlet nozzle and said device cover are detachably connected through said first fixing groove, said second fixing groove and said fixing ring.

12. The dumping-type automatic quantitative liquid discharging device according to claim 7, wherein a second valve body is arranged in said second liquid storage cavity, and said second valve body is rollably arranged in said second liquid storage cavity and closely adheres to said second liquid storage cavity for blocking.

13. The dumping-type automatic quantitative liquid discharging device according to claim 6, wherein an air inlet pipe is arranged on said device, and said air inlet pipe is communicated with said fluid channel.

14. The dumping-type automatic quantitative liquid discharging device according to claim 7,

wherein a second clamping block is arranged on said second liquid storage cavity, and a second clamping groove corresponding to said second clamping block is arranged on said first liquid storage cavity, and said first liquid storage cavity and said second liquid storage cavity are detachably connected by clamping.

15. The dumping-type automatic quantitative liquid discharging device according to claim 10, wherein a handle part is arranged on said device cover.

16. The dumping-type automatic quantitative liquid discharging device according to claim 6, wherein said accommodating cavity is provided with anti-skid lines.

17. The dumping-type automatic quantitative liquid discharging device according to claim 7, wherein at least one limiting edge corresponding to said first valve body is arranged in said fluid channel.

18. The dumping-type automatic quantitative liquid discharging device according to claim 10, wherein external threads are arranged on said accommodating cavity and internal threads are arranged on said device cover, and said accommodating cavity and said device cover are detachably connected through threads.

19. The dumping-type automatic quantitative liquid discharging device according to claim 10, wherein a second sealing ring and a sealing groove corresponding to said second sealing ring are arranged on said device cover.

20. A quantitative liquid discharging method, comprising providing a dumping-type automatic quantitative liquid discharging device, wherein dumping-type automatic quantitative liquid discharging device comprises an accommodating cavity and a device cover for accommodating liquid, a first liquid storage cavity and a second liquid storage cavity for storing quantitative liquid, and a fluid channel; and wherein, said fluid channel sequentially communicates with said accommodating cavity, said first liquid storage cavity and said second liquid storage cavity, and defines a path for air and said quantitative liquid to enter and leave said device; and wherein, said first liquid storage cavity and said second liquid storage cavity are arranged at one end of said accommodating cavity; and when said device pours out liquid, the quantitative liquid stored in said second liquid storage cavity leaves said device through said fluid channel, and an atmospheric low pressure area is formed at the other end of said accommodating cavity; and an atmospheric normal pressure area is formed in said first liquid storage cavity and said second liquid storage cavity, so that said liquid is confined between said first liquid storage cavity and said accommodating cavity; and wherein said second liquid storage cavity captures and stores a part of said liquid to become new said quantitative liquid; and wherein, when said device is reset, said quantitative liquid stored in said second liquid storage cavity will be received and stored by said first liquid storage cavity; and pouring liquid into said accommodating cavity; and connecting said device cover with said accommodating cavity; and tilting said dumping-type automatic quantitative liquid discharging device to allow the quantitative liquid to flow out.
